

## Estimation Accuracy

Empirical Bayes methods blur the line between estimation and hypothesis testing:  $\widehat{\text{fdr}}(z_i)$  (5.5) estimates the probability that a null hypothesis is correct, and likewise  $\widehat{\text{Fdr}}(z_i)$  (5.6). How accurate are such estimates, like the ones shown in Figure 5.2? In general we would like to assess the accuracy of summary statistics  $s(\mathbf{z})$  that depend on a vector of  $z$ -values  $\mathbf{z} = (z_1, z_2, \dots, z_N)$ . This would be straightforward if the  $z_i$  were independent, using classical methods or perhaps the bootstrap.

Independence is only a dream, though, for most large-scale data sets. Microarray studies are particularly prone to correlation across cases, as will be seen in Chapter 8. Now it seems we are in real trouble, where we must estimate an  $N \times N$  matrix of correlations in order to assess a standard deviation for  $s(\mathbf{z})$ . Fortunately, the main result of this chapter shows that this isn't necessary: under reasonable assumptions, good accuracy approximations are available that depend only upon the root mean square of the  $N \cdot (N - 1)/2$  correlations, a quantity that often can be easily estimated.

As an example of these results, suppose  $s(\mathbf{z})$  is the right-sided cdf for the leukemia data of Figure 6.1a,

$$\bar{F}(x) = \#\{z_i \geq x\}/N \quad [N = 7128]. \quad (7.1)$$

Section 7.1 and Section 7.2 show that a good approximation for the variance<sup>1</sup> of  $\bar{F}(x)$  is

$$\text{var}\{\bar{F}(x)\} \doteq \left\{ \frac{\bar{F}(x) \cdot (1 - \bar{F}(x))}{N} \right\} + \left\{ \frac{\hat{\alpha} \hat{\sigma}_0^2 \hat{f}^{(1)}(x)}{\sqrt{2}} \right\}^2 \quad (7.2)$$

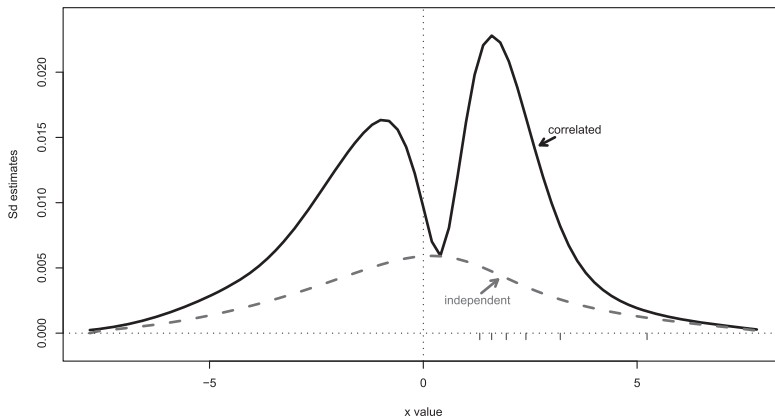
where

- $\hat{\alpha}$  is the estimated root mean square correlation (0.11 for the leukemia data, Chapter 8);

<sup>1</sup> Equation (7.50) gives the full covariance matrix for  $\bar{F}(x_k)$ ,  $k = 1, 2, \dots, K$ .

- $\hat{\sigma}_0$  is the estimated central histogram spread (1.68 for the leukemia data, Figure 6.1a);
- $\hat{f}^{(1)}(x)$  is the first derivative of the Poisson regression estimate of the mixture density  $f(z)$ , as derived in Section 5.2.

The first term in (7.2) is the usual binomial variance, while the second term is a *correlation penalty* accounting for dependence between the  $z_i$  values.



**Figure 7.1** Solid curve is estimated standard deviation of right-sided cdf (7.1) for leukemia  $z$ -values, from formula (7.2). Dashed curve shows binomial standard deviation from first term of (7.2), ignoring the correlation penalty. Correlation greatly increases the variability of  $\bar{F}(x)$  in this case. Hash marks indicate right percentiles used in Table 7.1.

The heavy curve in Figure 7.1 graphs estimated standard deviations  $\text{sd}\{\bar{F}(x)\}$  for the leukemia data, the square root of formula (7.2). For comparison, the dashed curve shows the usual binomial standard deviation estimates that would apply if the  $z_i$  were independent, the square root of the first term in (7.2). The correlation penalty can be enormous here, multiplying standard deviations by large factors, as shown numerically in Table 7.1.

This chapter derives accuracy formulas like (7.2) for a class of summary statistics  $s(z)$  pertaining to empirical Bayes calculations, including those seen in Figure 5.3. Our main assumption is that the  $z_i$  are individually normal with possibly different means and variances,

$$z_i \sim \mathcal{N}(\mu_i, \sigma_i^2) \quad \text{for } i = 1, 2, \dots, N. \quad (7.3)$$

Table 7.1 *Bold figures are estimated standard deviations for right-sided cdf (7.1), leukemia z-values, formula (7.2);  $sd_{ind}$  are binomial standard deviations, ignoring z-value correlations. Standard deviations evaluated at indicated right percentiles of the  $N = 7128$   $z_i$ 's.*

| Percentile | .25         | .20         | .15         | .10         | .05          | .01          |
|------------|-------------|-------------|-------------|-------------|--------------|--------------|
| <b>sd</b>  | <b>.021</b> | <b>.023</b> | <b>.021</b> | <b>.016</b> | <b>.0082</b> | <b>.0017</b> |
| $sd_{ind}$ | .005        | .005        | .004        | .004        | .0026        | .0012        |
| ratio      | 4.1         | 4.8         | 5.0         | 4.6         | 3.2          | 1.4          |

There is no requirement that they be “z-values” in the hypothesis-testing sense (2.6), though that is our main class of applications.

It will take us some time to justify formula (7.2) and its extensions. The argument proceeds as follows:

- Exact cdf covariance formulas are derived in Section 7.1.
- Simplified approximations are developed in Section 7.2, along with practical data-based estimates.
- Delta-method arguments are used to extend the cdf results to more general summary statistics in Section 7.3.
- Section 7.4 concerns the non-null distribution of z-values, showing model (7.3) to be a good approximation and justifying application of the theory.

The calculations and derivations, some of which are quite involved, emphasize the frequentist side of empirical Bayes theory. Technical details, particularly in Section 7.1 and Section 7.4, can be bypassed for quick reading once the definitions and results are understood, the figures being helpful in this regard.

A natural question is why not use the bootstrap to get accuracy estimates directly? Section 7.5 provides a brief answer: with  $N$  much greater than  $n$ , the non-parametric bootstrap can behave eccentrically. Figure 7.8 shows this happening in the context of Figure 7.1.

7.1 Exact Covariance Formulas

Given correlated normal observations  $z_1, z_2, \dots, z_N$  (7.3), this section derives exact formulas for the mean and covariance of the empirical process  $\{\bar{F}(x), -\infty < x < \infty\}$ , that is, for the right-sided cdf (7.1) evaluated for all choices of  $x$ . Rather than work directly with the cdf, it will be easier to derive first our results for a discretized version of the empirical *density* of the

$z_i$  values; more precisely, we will compute the mean vector and covariance matrix for the histogram counts  $y_k$  (5.12). Most statistics  $s(\mathbf{z})$  of interest to us depend on  $\mathbf{z}$  only through the vector of counts.

The notation here will be that of (5.11)–(5.14):  $y_k$  is the number of  $z_i$  values in bin  $k$ ,  $k = 1, 2, \dots, K$ , giving count vector

$$\mathbf{y} = (y_1, y_2, \dots, y_K)'; \quad (7.4)$$

$x_k$  is the centerpoint of bin  $k$ ,  $\mathbf{x} = (x_1, x_2, \dots, x_K)'$ ; and all bins are of width  $d$ .

**Exercise 7.1** Let  $z_{\text{ord}}$  be the order statistic of  $\mathbf{z}$ , that is, the ordered values  $z_{(1)} < z_{(2)} < \dots < z_{(N)}$  (assuming no ties). Show that, for sufficiently small choice of the bin width  $d$ , we can recover  $z_{\text{ord}}$  from  $\mathbf{y}$ .

*Note* All of our previous estimators, such as  $\widehat{\text{fdr}}(\mathbf{z})$  and  $\widehat{\text{Fdr}}(\mathbf{z})$ , depend on  $\mathbf{z}$  only through  $z_{\text{ord}}$  and therefore, effectively, on  $\mathbf{y}$ .

Instead of (7.3), we first make the more restrictive assumption that the  $z_i$  are divided into a finite number of classes, with members of the  $c$ th class  $C_c$  having mean  $\mu_c$  and standard deviation  $\sigma_c$ ,

$$z_i \sim \mathcal{N}(\mu_c, \sigma_c^2) \quad \text{for } z_i \in C_c. \quad (7.5)$$

Let  $N_c$  be the number of members of  $C_c$ , with  $\pi_c$  the proportion,

$$N_c = \#C_c \quad \text{and} \quad \pi_c = N_c/N \quad (7.6)$$

so  $\sum_c N_c = N$  and  $\sum_c \pi_c = 1$ .

It will be convenient to define *adjusted bin centers*  $x_{kc} = (x_k - \mu_c)/\sigma_c$  for  $k = 1, 2, \dots, K$ , denoting the whole  $K$ -vector by

$$\mathbf{x}_c = (x_{1c}, x_{2c}, \dots, x_{Kc})'. \quad (7.7)$$

Similarly, if  $h(x)$  is any function of  $x$ , we will write

$$\mathbf{h}_c = h(\mathbf{x}_c) = (h(x_{1c}), h(x_{2c}), \dots, h(x_{Kc}))' \quad (7.8)$$

for  $h$  evaluated at the adjusted bin centers; so, for example,  $\varphi_c$  is the standard normal density (2.9) evaluated over  $\mathbf{x}_c$ .

It is easy to calculate the expectation of the count vector  $\mathbf{y}$  under model (7.5)–(7.6). Let  $P_{kc}$  be the probability that  $z_i$  from class  $C_c$  falls into the  $k$ th bin  $\mathcal{Z}_k$ ,

$$P_{kc} = \Pr\{z_i \in \mathcal{Z}_k\} \doteq d \cdot \varphi(x_{kc})/\sigma_c. \quad (7.9)$$

This last approximation becomes arbitrarily accurate for  $d$  sufficiently small, and we will take it to be exact in what follows. Then

$$E\{\mathbf{y}\} = N \sum_c \pi_c \mathbf{P}_c = Nd \sum_c \pi_c \boldsymbol{\varphi}_c, \quad (7.10)$$

$\mathbf{P}_c = (P_{1c}, P_{2c}, \dots, P_{Kc})'$  as in (7.8).

**Exercise 7.2** Verify (7.10).

The  $K \times K$  covariance matrix of the count vector  $\mathbf{y}$  depends on the  $N \times N$  correlation matrix of  $\mathbf{z}$ , but in a reasonably simple way discussed next. Two important definitions are needed to state the first result: there are  $M = N(N-1)/2$  correlations  $\rho_{ii'}$  between pairs  $(z_i, z_{i'})$  of members of  $\mathbf{z}$ , and we denote by  $g(\rho)$  the distribution<sup>2</sup> putting weight  $1/M$  on each  $\rho_{ii'}$ . Also, for  $\varphi_\rho(u, v)$ , the bivariate normal density having zero means, unit standard deviations, and correlation  $\rho$ , we define

$$\begin{aligned} \lambda_\rho(u, v) &= \frac{\varphi_\rho(u, v)}{\varphi(u)\varphi(v)} - 1 \\ &= (1 - \rho^2)^{-\frac{1}{2}} \exp \left\{ \frac{2\rho uv - \rho^2(u^2 + v^2)}{2(1 - \rho^2)} \right\} - 1 \end{aligned} \quad (7.11)$$

and

$$\lambda(u, v) = \int_{-1}^1 \lambda_\rho(u, v) g(\rho) d\rho. \quad (7.12)$$

**Lemma 7.1** Under the multi-class model (7.5)–(7.6), the covariance of the count vector  $\mathbf{y}$  (7.4) has two components,

$$\mathbf{cov}(\mathbf{y}) = \mathbf{cov}_0 + \mathbf{cov}_1 \quad (7.13)$$

where

$$\mathbf{cov}_0 = N \sum_c \pi_c \{\text{diag}(\mathbf{P}_c) - \mathbf{P}_c \mathbf{P}_c'\} \quad (7.14)$$

and

$$\begin{aligned} \mathbf{cov}_1 &= N^2 \sum_c \sum_d \pi_c \pi_d \text{diag}(\mathbf{P}_c) \boldsymbol{\lambda}_{cd} \text{diag}(\mathbf{P}_d) \\ &\quad - N \sum_c \text{diag}(\mathbf{P}_c) \boldsymbol{\lambda}_{cc} \text{diag}(\mathbf{P}_c). \end{aligned} \quad (7.15)$$

<sup>2</sup>  $g(\rho)$  is a discrete distribution but we will treat it notationally as a density function.

Formula (7.17), for example, indicates the average of  $\rho^j$  over the  $M$  values  $\rho_{ii'}$ .

Here  $\text{diag}(\mathbf{P}_c)$  is the  $K \times K$  diagonal matrix having diagonal elements  $P_{kc}$ , similarly  $\text{diag}(\mathbf{P}_d)$ , while  $\lambda_{cd}$  is the  $K \times K$  matrix with  $kl$ th element  $\lambda(x_{kc}, x_{ld})$  (7.12). Summations are over all classes.<sup>3</sup> The proof of Lemma 7.1 appears at the end of this section.

*Note* Equation (7.15) assumes that the correlation distribution  $g(\rho)$  is the same across all classes  $C_c$ .

The  $\mathbf{cov}_0$  term in (7.13)–(7.14) is the sum of the multinomial covariance matrices that would apply if the  $z_i$  were mutually independent with fixed numbers drawn from each class;  $\mathbf{cov}_1$  is the correlation penalty, almost always increasing  $\mathbf{cov}(\mathbf{y})$ . The  $N^2$  factor in (7.15) makes the correlation penalty more severe as  $N$  increases, assuming  $g(\rho)$  stays the same.

Expression (7.15) for the correlation penalty can be considerably simplified. *Mehler's identity*<sup>4</sup> for  $\lambda_\rho(u, v)$  (7.11) is

$$\lambda_\rho(u, v) = \sum_{j \geq 1} \frac{\rho^j}{j!} h_j(u) h_j(v) \quad (7.16)$$

where  $h_j$  is the  $j$ th Hermite polynomial. Denoting the  $j$ th moment of the correlation distribution  $g(\rho)$  by  $\alpha_j$ ,

$$\alpha_j = \int_{-1}^1 \rho^j g(\rho) d\rho, \quad (7.17)$$

(7.12) becomes

$$\lambda(u, v) = \sum_{j \geq 1} \frac{\alpha_j}{j!} h_j(u) h_j(v) \quad (7.18)$$

so  $\lambda_{cd}$  in (7.15) can be expressed in outer product notation as

$$\lambda_{cd} = \sum_{j \geq 1} \frac{\alpha_j}{j!} h_j(\mathbf{x}_c) h_j(\mathbf{x}_d)'. \quad (7.19)$$

(The outer product of vectors  $\mathbf{u}$  and  $\mathbf{v}$  is the matrix having  $ij$ th element  $u_i v_j$ .) Making use of (7.9), taken as exact,

$$\begin{aligned} \text{diag}(\mathbf{P}_c) h_j(\mathbf{x}_c) &= Nd \text{diag}(\varphi(\mathbf{x}_c)) h_j(\mathbf{x}_c) / \sigma_c \\ &= (-1)^j Nd \varphi_c^{(j)} / \sigma_c \end{aligned} \quad (7.20)$$

where  $\varphi_c^{(j)}$  indicates the  $j$ th derivative of  $\varphi(u)$  evaluated at each component

<sup>3</sup> Note that subscript  $d$  is distinct from bin width  $d$ , which is a constant.

<sup>4</sup> Equation (7.16) is also known as the *tetrachoric series* and has interesting connections to canonical correlation, the singular value decomposition, Pearson's coefficient of contingency, and correspondence analysis.

of  $\mathbf{x}_c$ . The last line of (7.20) uses the Hermite polynomial relationship  $\varphi^{(j)}(u) = (-1)^j \varphi(u) h_j(u)$ .

Rearranging (7.15) then gives a simplified formula for  $\mathbf{cov}_1$ .

**Lemma 7.2** *Defining*

$$\bar{\phi}^{(j)} = \sum_c \pi_c \varphi_c^{(j)} / \sigma_c, \quad (7.21)$$

formula (7.15) for the correlation penalty becomes

$$\mathbf{cov}_1 = N^2 d^2 \left\{ \sum_{j \geq 1} \frac{\alpha_j}{j!} \bar{\phi}^{(j)} \bar{\phi}^{(j)'} - \frac{1}{N} \sum_{j \geq 1} \frac{\alpha_j}{j!} \left( \sum_c \pi_c \varphi_c^{(j)} \varphi_c^{(j)'} / \sigma_c^2 \right) \right\}. \quad (7.22)$$

Returning to right-sided cdfs (7.1), let  $\mathbf{B}$  be the  $K \times K$  matrix

$$B_{kk'} = \begin{cases} 1 & \text{if } k \leq k' \\ 0 & \text{if } k > k' \end{cases} \quad (7.23)$$

so

$$\bar{\mathbf{F}} = \frac{1}{N} \mathbf{B} \mathbf{y} \quad (7.24)$$

is a  $K$ -vector with  $k$ th component the proportion of  $z_i$  values in bins indexed  $\geq k$ ,

$$\bar{F}_k = \#\{z_i \geq x_k - d/2\} / N \quad (k = 1, 2, \dots, K). \quad (7.25)$$

( $\mathbf{B}$  would be transposed if we were dealing with left-sided cdfs.)

The expectation of  $\bar{\mathbf{F}}$  is easy to obtain from (7.10),

$$\begin{aligned} E\{\bar{F}_k\} &= \sum_c \pi_c \left[ \sum_{k' \geq k} d \varphi \left( \frac{x_{k'} - \mu_c}{\sigma_c} \right) / \sigma_c \right] \doteq \sum_c \pi_c \int_{x_{kc}}^{\infty} \varphi(u) du \\ &= \sum_c \pi_c \Phi^+(x_{kc}) \end{aligned} \quad (7.26)$$

where  $\Phi^+(u) \equiv 1 - \Phi(u)$ . Now that we are working with tail areas rather than densities, we can let the bin width  $d \rightarrow 0$ , making (7.26) exact.

$\bar{\mathbf{F}}$  has covariance matrix  $\mathbf{B} \mathbf{cov}(\mathbf{y}) \mathbf{B}' / N^2$ . Applying integration calculations like that in (7.26) to Lemma 7.1 gives a covariance decomposition for  $\bar{\mathbf{F}}$ .

**Theorem 7.3** *Under the multi-class model (7.5)–(7.6),*

$$\mathbf{Cov}(\bar{\mathbf{F}}) = \mathbf{Cov}_0 + \mathbf{Cov}_1 \quad (7.27)$$

where  $\mathbf{Cov}_0$  has  $kl$ th entry

$$\frac{1}{N} \sum_c \pi_c \{ \Phi^+ (\max(x_{kc}, x_{lc})) - \Phi^+(x_{kc})\Phi^+(x_{lc}) \} \quad (7.28)$$

and

$$\mathbf{Cov}_1 = \sum_j \frac{\alpha_j}{j!} \bar{\varphi}^{(j-1)} \bar{\varphi}^{(j-1)'} - \frac{1}{N} \sum_j \frac{\alpha_j}{j!} \left\{ \sum_c \pi_c \bar{\varphi}_c^{(j-1)} \bar{\varphi}_c^{(j-1)'} \right\}. \quad (7.29)$$

Here  $\pi_c$  is from (7.6),  $x_{kc}$  and  $x_{lc}$  are from (7.7),  $\alpha_j$  is as in (7.17), and

$$\bar{\varphi}^{(j-1)} = \sum_c \pi_c \varphi_c^{(j-1)}. \quad (7.30)$$

(Note the distinction between  $\bar{\varphi}$  and  $\bar{\phi}$  (7.21).)

*Proof of Lemma 7.1* Let  $I_k(i)$  denote the indicator function of the event  $z_i \in \mathcal{Z}_k$  (5.11) so that the number of  $z_i$  values from class  $C_c$  in  $\mathcal{Z}_k$  is

$$y_{kc} = \sum_i I_k(i), \quad (7.31)$$

the boldface subscript indicating summation over the members of  $C_c$ . We first compute  $E\{y_{kc}y_{ld}\}$  for bins  $k$  and  $l$ ,  $k \neq l$ , and classes  $c$  and  $d$ ,

$$\begin{aligned} E\{y_{kc}y_{ld}\} &= E \left\{ \sum_c \sum_d I_k(i) I_l(j) \right\} \\ &= d^2 \sum_c \sum_d \frac{\varphi_{\rho_{ij}}(x_{kc}, x_{ld})(1 - \chi_{ij})}{\sigma_c \sigma_d} \end{aligned} \quad (7.32)$$

following notation (7.6)–(7.10), with  $\chi_{ij}$  the indicator function of event  $i = j$  (which can only occur if  $c = d$ ). This reduces to

$$E\{y_{kc}y_{ld}\} = N^2 d^2 \pi_c (\pi_d - \chi_{cd}/N) \frac{\int_{-1}^1 \varphi_\rho(x_{kc}, x_{ld}) g(\rho) d\rho}{\sigma_c \sigma_d} \quad (7.33)$$

under the assumption that the same correlation distribution  $g(\rho)$  applies across all class combinations. Since  $y_k = \sum_c y_{kc}$ , we obtain

$$E\{y_k y_l\} = N^2 d^2 \sum_c \sum_d \pi_c (\pi_d - \chi_{cd}/N) \frac{\int_{-1}^1 \varphi_\rho(x_{kc}, x_{ld}) g(\rho) d\rho}{\sigma_c \sigma_d}, \quad (7.34)$$

the non-bold subscripts indicating summation over classes. Subtracting

$$E\{y_k\}E\{y_l\} = N^2 d^2 \sum_c \sum_d \frac{\varphi(x_{kc})\varphi(x_{ld})}{\sigma_c \sigma_d} \quad (7.35)$$

from (7.34) results, after some rearrangement, in

$$\begin{aligned} \text{cov}(y_k, y_l) = & N^2 d^2 \sum_c \sum_d \frac{\varphi(x_{kc})\varphi(x_{ld})}{\sigma_c \sigma_d} \\ & \times \left\{ \pi_c \left( \pi_d - \frac{\chi_{cd}}{N} \right) \int_{-1}^1 \left( \frac{\varphi_\rho(x_{kc}, x_{ld})}{\varphi(x_{kc})\varphi(x_{ld})} - 1 \right) g(\rho) d\rho \right\} \\ & - N d^2 \sum_c \pi_c \frac{\varphi(x_{kc})\varphi(x_{ld})}{\sigma_c \sigma_d}. \end{aligned} \quad (7.36)$$

Using  $P_{kc} = d \cdot \varphi(x_{kc})/\sigma_c$  as in (7.9), expression (7.36) is seen to equal the  $k$ th element of  $\mathbf{cov}(\mathbf{y})$  in Lemma 7.1, when  $k \neq l$ . The case  $k = l$  proceeds in the same way, the only difference being that  $N d \pi_c \chi_{cd} \varphi(x_{kc})/\sigma_c$  must be added to formula (7.32), again in agreement with  $\mathbf{cov}(\mathbf{y})$  in Lemma 7.1.  $\square$

## 7.2 Rms Approximations

The exact covariance expressions of Section 7.1 are useful for theoretical and simulation calculations, but less so for actual applications. This section develops simplified versions called *rms approximations* (for “root mean square”) that facilitate the kind of data-based estimates seen in (7.2).

Convenient approximations to  $\mathbf{Cov}_1$  and  $\mathbf{cov}_1$  are based on three simplifications:

- 1 The second terms in (7.28) and in (7.22) are of order  $1/N$  compared to the first terms, and can be ignored for  $N$  large.
- 2 Common standardization methods for data matrix  $\mathbf{X}$ , such as subtracting off the mean of each column as we did in our microarray examples, make  $\alpha_1$ , the first coefficient in Mehler’s identity (7.18), exactly or nearly zero; this leaves  $\alpha_2$  as the lead.
- 3 With  $\rho$  confined to  $[-1, 1]$ , the higher-order moments  $\alpha_j$  of  $g(\rho)$ ,  $j \geq 3$ , decrease quickly to zero if  $\alpha_2$  is not too large.

Making all three simplifications reduces (7.28) and (7.22) to the rms approximations

$$\mathbf{Cov}_1 \doteq \alpha^2 \bar{\varphi}^{(1)} \bar{\varphi}^{(1)'} / 2 \quad (7.37)$$

and

$$\mathbf{cov}_1 \doteq (N d \alpha)^2 \bar{\phi}^{(2)} \bar{\phi}^{(2)'} / 2 \quad (7.38)$$

where  $\alpha$  is the *root mean square correlation*

$$\alpha = \alpha_2^{1/2} = \left[ \int_{-1}^1 \rho^2 g(\rho) d\rho \right]^{1/2}. \quad (7.39)$$

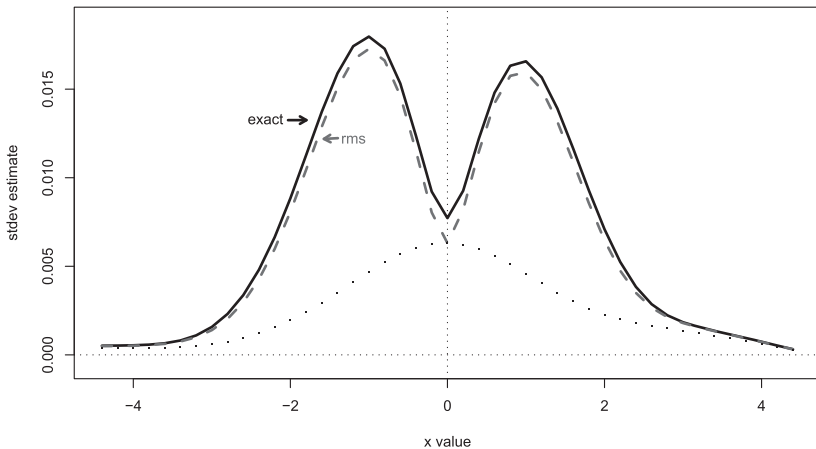
Here  $\bar{\varphi}^{(1)}$  is the average first derivative in (7.30),

$$\bar{\varphi}^{(1)} = \sum_c \pi_c \varphi_c^{(1)} \quad (7.40)$$

and likewise

$$\bar{\phi}^{(2)} = \sum_c \pi_c \bar{\varphi}^{(2)} / \sigma_c \quad (7.41)$$

from (7.21).



**Figure 7.2** Comparison of exact formula for  $\text{sd}\{\bar{F}(x)\}$  from Theorem 7.3 (solid curve) with the rms approximation (dashed) for  $N = 6000$ ,  $\alpha = 0.10$ , in the two-class situation (7.42). Dots indicate binomial standard deviation, ignoring correlation penalty.

Figure 7.2 compares exact and approximate standard deviations of  $\bar{F}(x)$  (7.1), for the situation where  $N = 6000$ ,  $\alpha = 0.10$ , and there are two classes (7.5), (7.6) having

$$\begin{aligned} (\mu_0, \sigma_0) &= (-0.125, 1) & \pi_0 &= 0.95 \\ \text{and } (\mu_1, \sigma_1) &= (2.38, 1) & \pi_1 &= 0.05. \end{aligned} \quad (7.42)$$

(If we begin with  $\mu_0 = 0$  and  $\mu_1 = 2.5$  and recenter  $z$  by subtracting  $\bar{z}$ ,

we get  $\mu_0$  and  $\mu_1$  as in (7.42).) The solid curve traces the exact standard deviations, i.e., square roots of the diagonal elements of  $\mathbf{Cov}(\bar{F})$  (7.27); the dashed curve shows standard deviations when substituting the rms approximation (7.37) for  $\mathbf{Cov}_1$  in (7.27). We see that the rms approximation performs quite satisfactorily.

Practical application of the rms approximations requires us to estimate the rms correlation  $\alpha$  (7.39) and  $\bar{\varphi}^{(1)}$  (7.40) or  $\bar{\phi}^{(2)}$  (7.41). Chapter 8 shows that an estimate  $\hat{\alpha}$  is easy to obtain from data matrices  $\mathbf{X}$  such as that for the leukemia study. Here we will describe the estimation of  $\bar{\varphi}^{(1)}$  or  $\bar{\phi}^{(2)}$ . These look difficult since they depend on the class parameters  $(\mu_c, \sigma_c, \pi_c)$  in (7.5)–(7.6) but, at least under some assumptions, a simple estimate is available.

The marginal density  $f(z)$  under model (7.5)–(7.6) is

$$f(z) = \sum_c \pi_c \varphi\left(\frac{z - \mu_c}{\sigma_c}\right) \bigg| \sigma_c \quad (7.43)$$

so the first and second derivatives of  $f(z)$  are

$$\begin{aligned} f^{(1)}(z) &= \sum_c \pi_c \varphi^{(1)}\left(\frac{z - \mu_c}{\sigma_c}\right) \bigg| \sigma_c^2 \\ \text{and } f^{(2)}(z) &= \sum_c \pi_c \varphi^{(2)}\left(\frac{z - \mu_c}{\sigma_c}\right) \bigg| \sigma_c^3. \end{aligned} \quad (7.44)$$

Suppose we make the *homogeneity assumption* that all  $\sigma_c$  values are the same, say  $\sigma_c = \sigma_0$ . Comparison with definitions (7.30) and (7.21) then gives

$$\bar{\varphi}^{(1)} = \sigma_0^2 \mathbf{f}^{(1)} \quad \text{and} \quad \bar{\phi}^{(2)} = \sigma_0^2 \mathbf{f}^{(2)} \quad (7.45)$$

where  $\mathbf{f}^{(j)} = f^{(j)}(\mathbf{x})$ , the  $j$ th derivative evaluated at the bin centerpoints. It is worth noting that the class structure (7.5)–(7.6) has disappeared entirely in (7.45), leaving  $z_i \sim \mathcal{N}(\mu_i, \sigma_0^2)$  for  $i = 1, 2, \dots, N$  as the basic assumption.

**Exercise 7.3** Verify (7.45).

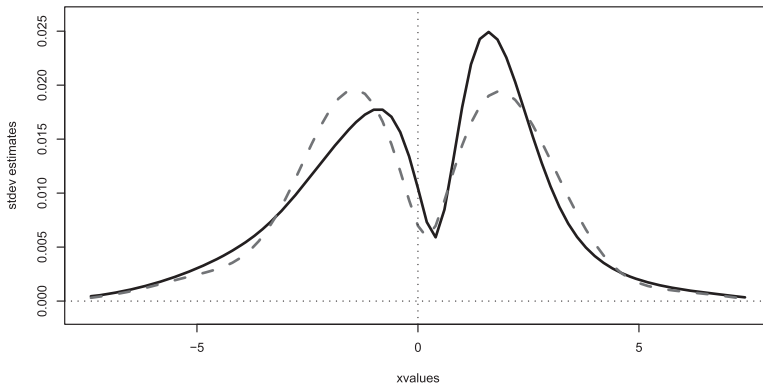
Substituting (7.45) into the rms approximations (7.37)–(7.38) yields convenient expressions for the correlation penalties,

$$\mathbf{Cov}_1 \doteq \frac{(\sigma_0^2 \alpha)^2}{2} \mathbf{f}^{(1)} \mathbf{f}^{(1)'} \quad \text{and} \quad \mathbf{cov}_1 \doteq \frac{(Nd\sigma_0^2 \alpha)^2}{2} \mathbf{f}^{(2)} \mathbf{f}^{(2)'} \quad (7.46)$$

This finally returns us to the correlation penalty used in equation (7.2),  $(\hat{\alpha} \hat{\sigma}_0^2 \hat{f}^{(1)}(x) / \sqrt{2})^2$ , where  $f^{(1)}(z)$  has been estimated by differentiation of the Poisson regression estimate  $\hat{f}(z)$  of Section 5.2.

Table 7.2 *Estimated 3-class model for the leukemia data.*

|            | Left | Center | Right |
|------------|------|--------|-------|
| $\mu_c$    | -4.2 | .09    | 5.4   |
| $\sigma_c$ | 1.16 | 1.68   | 1.05  |
| $\pi_c$    | .054 | .930   | .016  |



**Figure 7.3** Leukemia data; comparison of standard deviation estimates for empirical cdf (7.1), using formula (7.2) (solid curve) or 3-class model based on Table 7.2 (dashed curve).

Suppose we distrust the homogeneity assumption. The smoothed non-null counts  $\hat{y}_{1k}$  (5.42) allow direct, though rather crude, estimates of the class parameters  $(\mu_c, \sigma_c, \pi_c)$ . The equivalent of Figure 5.5 for the leukemia data, along with the MLE estimates  $(\hat{\mu}_0, \hat{\sigma}_0, \hat{\pi}_0)$ , suggests a 3-class model with the parameter values given in Table 7.2. The dashed curve in Figure 7.3 shows standard deviations for  $\bar{F}(x)$  if  $\hat{\varphi}^{(1)}$  in (7.29) is estimated from (7.40) using the parameter values in Table 7.2. It differs only modestly from the results based on the homogeneity assumption (7.45), as used in formula (7.2).

**Exercise 7.4** How were the left and right class parameters in Table 7.2 calculated from the smooth counts  $\hat{y}_{1k}$ ?

The independence terms as stated in Theorem 7.3 and Lemma 7.1,  $\mathbf{Cov}_0$  and  $\mathbf{cov}_0$ , depend on the class structure (7.5)–(7.6). They assume that the class sample sizes  $N_c$  are fixed constants so that  $\mathbf{cov}_0$  is a mixture of multi-

nomial covariance matrices. A more realistic assumption might be that  $N_1, N_2, \dots, N_C$  are a multinomial sample of size  $N$ , sampled with probabilities  $\pi_1, \pi_2, \dots, \pi_C$ , in which case  $\mathbf{cov}_0$  becomes the single multinomial covariance

$$\mathbf{cov}_0 = N \left\{ \text{diag}(d \cdot \mathbf{f}) - d^2 \mathbf{f} \mathbf{f}' \right\}, \quad (7.47)$$

$\mathbf{f} = f(\mathbf{x})$  (7.43). Here  $d \cdot \mathbf{f}$  is the vector of bin probabilities

$$d \cdot \mathbf{f} = \sum_c \pi_c P_c \quad (7.48)$$

as in (7.9)–(7.10).

**Exercise 7.5** (a) Verify (7.48). (b) Show that  $\mathbf{cov}_0$  (7.47) is greater than or equal to  $\mathbf{cov}_0$  (7.14) in the sense that the difference is a positive semi-definite matrix.

Formula (7.47) has the advantage of not depending explicitly on the class structure. In the same way, we can replace  $\mathbf{Cov}_0$  (7.28) with a single multinomial expression.

To summarize, the rms approximations and the homogeneity assumption together yield these convenient estimates for the covariance matrices of the counts and their cdf values:

$$\widehat{\mathbf{cov}}\{\mathbf{y}\} = N \left\{ \text{diag}(d \cdot \hat{\mathbf{f}}) - d^2 \hat{\mathbf{f}} \hat{\mathbf{f}}' \right\} + \frac{(\hat{\alpha} N d \hat{\sigma}_0^2)^2}{2} \hat{\mathbf{f}}^{(2)} \hat{\mathbf{f}}^{(2)'} \quad (7.49)$$

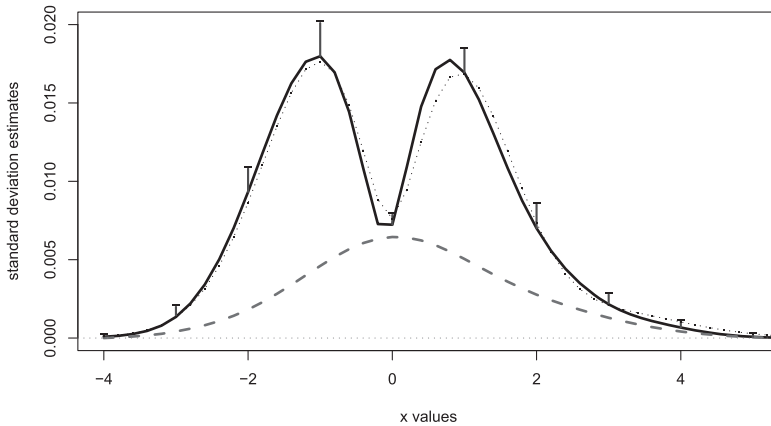
and

$$\widehat{\mathbf{Cov}}\{\bar{\mathbf{F}}\} = \frac{1}{N} \left\{ (\bar{F}_{\max(k,l)} - \bar{F}_k \bar{F}_l) \right\} + \frac{(\hat{\alpha} \hat{\sigma}_0^2)^2}{2} \hat{\mathbf{f}}^{(1)} \hat{\mathbf{f}}^{(1)'} \quad (7.50)$$

Here  $\bar{\mathbf{F}} = (\bar{F}(x_1), \bar{F}(x_2), \dots, \bar{F}(x_K))'$ , the leading matrix having elements as indicated in (7.50). Expression (7.2) is obtained from the diagonal elements of (7.50).

**Exercise 7.6** Show that the “independence” matrices in (7.49)–(7.50) are related by  $\widehat{\mathbf{Cov}}_0 = \mathbf{B} \widehat{\mathbf{cov}}_0 \mathbf{B}'$ , with  $\mathbf{B}$  as in (7.23).

Formula (7.2) for the standard deviation of  $\bar{F}(x)$  was tested in a simulation experiment: for each of 100 replications, a  $6000 \times 80$  matrix  $\mathbf{X}$  was constructed such that two-sample  $t$ -statistics comparing the first and last 40 columns gave  $z$ -values satisfying model (7.42), with  $N = 6000$  and  $\alpha = 0.10$  (see Section 8.2). The solid curve in Figure 7.4 graphs the mean



**Figure 7.4** Simulation experiment testing formula (7.2), 100 replications of model (7.42),  $N = 6000$ ,  $\alpha = 0.10$ , generated as described in Chapter 8. *Solid curve* is mean of 100 values  $\widehat{sd}_k$ , square root of (7.2) at  $x = x_k$ ; error bars indicate empirical standard deviations of  $\widehat{sd}_k$  at  $x = -4, -3, \dots, 4$ . *Dotted curve*: true  $sd_k$  from Theorem 7.3. *Dashed curve* is mean of  $\widehat{sd}_0$ , square root of first term in (7.2).

of the 100 values of  $\widehat{sd}_k$  (square roots of (7.2) for  $x = x_k$ ) with error bars indicating the empirical standard deviations. The dotted curve shows true  $sd_k$  values from Theorem 7.3. We see that formula (7.2) gave nearly unbiased results except at the extreme right, with reasonably low variability.

### 7.3 Accuracy Calculations for General Statistics

We would like to assess the accuracy of estimates such as  $\widehat{fdr}(z)$  and  $\widehat{Fdr}(z)$ . To this end, standard *delta-method* (i.e., Taylor series) techniques allow us to extend the covariance estimates of Section 7.2 to statistics that can be expressed as smoothly differentiable functions of the count vector  $\mathbf{y}$  (7.4).

Suppose that  $Q(\mathbf{y})$  is a  $q$ -dimensional function of the  $K$ -vector  $\mathbf{y}$  (considered as varying continuously) such that a small change  $d\mathbf{y}$  produces change  $dQ$  in  $Q$  according to the local linear approximation

$$dQ = \hat{D}d\mathbf{y} \quad (7.51)$$

where  $\hat{D}$  is the  $q \times K$  matrix  $dQ/d\mathbf{y}$ , with  $jk$ th element

$$\hat{D}_{jk} = \left. \frac{\partial Q_j}{\partial y_k} \right|_{\mathbf{y}}. \quad (7.52)$$

If  $\widehat{\mathbf{cov}}(\mathbf{y})$  is a covariance estimate for  $\mathbf{y}$ , perhaps (7.49), then the usual delta-method estimate of  $\mathbf{cov}(Q)$  is the  $q \times q$  matrix

$$\widehat{\mathbf{cov}}(Q) = \hat{D} \widehat{\mathbf{cov}}(\mathbf{y}) \hat{D}'. \quad (7.53)$$

In a theoretical context where  $\mathbf{cov}(\mathbf{y})$  is known we could use instead

$$\mathbf{cov}(Q) \doteq D \mathbf{cov}(\mathbf{y}) D', \quad (7.54)$$

now with the derivative matrix  $D$  evaluated at the expectation of  $\mathbf{y}$ . Sometimes  $D$  or  $\hat{D}$  is called the *influence function* of  $Q(\mathbf{y})$ .

As a first example, consider assessing the accuracy of  $\widehat{\text{fdr}}(z) = \pi_0 \hat{f}_0(z) / \hat{f}(z)$  (5.5), except with  $\pi_0$  considered known. Notation (7.8) will be useful here, where, if  $h(z)$  is any function of  $z$ , then  $\mathbf{h} = h(\mathbf{x}) = (h(x_1), h(x_2), \dots, h(x_K))'$  is  $h(z)$  evaluated at the bin centers  $\mathbf{x}$ . Thus, for example,  $\widehat{\mathbf{fdr}} = \pi_0 \mathbf{f}_0 / \hat{\mathbf{f}}$  is the  $K$ -vector<sup>5</sup> of values  $\widehat{\text{fdr}}(x_k)$ .

It is convenient to work first with the logarithm of  $\widehat{\mathbf{fdr}}$ , say  $\widehat{\mathbf{l fdr}} = \log(\widehat{\mathbf{fdr}})$ ,

$$\widehat{\mathbf{l fdr}} = \log(\pi_0) + \mathbf{l}_0 - \hat{\mathbf{l}}. \quad (7.55)$$

Since  $\log(\pi_0)$  and  $\mathbf{l}_0 = \log(\mathbf{f}_0)$  are constants, the influence function  $d\widehat{\mathbf{l fdr}}/d\mathbf{y}$  equals  $-d\hat{\mathbf{l}}/d\mathbf{y}$ .

The influence function  $d\hat{\mathbf{l}}/d\mathbf{y}$  is well known when  $\hat{\mathbf{f}}$  is the Poisson regression estimate of Section 5.2. Let  $M$  be the  $K \times m$  structure matrix for the regression. (In the implementation of (5.10),  $M$  has  $k$ th row  $(1, x_k, x_k^2, \dots, x_k^J)$ ,  $m = J + 1$ .) Also let  $\boldsymbol{\nu} = N d\hat{\mathbf{f}}$  be the expectation of  $\mathbf{y}$  (5.14) with<sup>6</sup>  $\hat{\boldsymbol{\nu}} = N d\hat{\mathbf{f}}$  and

$$\hat{G}_{m \times m} = M' \text{diag}(\hat{\boldsymbol{\nu}}) M, \quad (7.56)$$

$\text{diag}(\hat{\boldsymbol{\nu}})$  indicating a  $K \times K$  diagonal matrix with diagonal elements  $\hat{\nu}_k$ . A standard generalized linear model (GLM) argument discussed at the end of this section gives

$$d\hat{\mathbf{l}}/d\mathbf{y} = M \hat{G} M' = -d\widehat{\mathbf{l fdr}}/d\mathbf{y}. \quad (7.57)$$

<sup>5</sup> Following conventions of computer language R, the ratio of vectors is the vector of ratios, etc., and  $\log(\pi_0)$  is added to each component at (7.55).

<sup>6</sup> The estimates  $\hat{\mathbf{f}}$  and  $\hat{\mathbf{f}}_0$  returned by `locfdr` sum to  $N$  rather than to  $1/d$  for easy graphical comparison with  $\mathbf{y}$ .

Applying (7.53) yields an estimated covariance matrix for  $\widehat{\mathbf{lfd}}r$ ,

$$\widehat{\mathbf{cov}}(\widehat{\mathbf{lfd}}r) = (M\hat{G}M') \widehat{\mathbf{cov}}(\mathbf{y}) (M\hat{G}M') \quad (7.58)$$

where  $\widehat{\mathbf{cov}}(\mathbf{y})$  can be taken from (7.49). If we are in the independence situation  $\hat{\alpha} = 0$ , then we can write (7.49) as

$$\widehat{\mathbf{cov}}(\mathbf{y}) = \text{diag}(\hat{\mathbf{v}}) - \hat{\mathbf{v}}\hat{\mathbf{v}}'/N. \quad (7.59)$$

Ignoring the last term, (7.58) gives a nice result,

$$\widehat{\mathbf{cov}}(\widehat{\mathbf{lfd}}r) \doteq M\hat{G}M' \text{diag}(\hat{\mathbf{v}})M\hat{G}M' = M\hat{G}M'. \quad (7.60)$$

In our examples, however, the correlation penalty was usually too big to ignore.

Corresponding to  $\widehat{\mathbf{lfd}}r$  (7.55) is the right-sided log tail area false discovery rate

$$\widehat{\mathbf{lfd}}r = \log(\pi_0) + \mathbf{L}_0 - \hat{\mathbf{L}} \quad (7.61)$$

where  $\hat{\mathbf{L}} = \log(\hat{F}(\mathbf{x}))$  and  $\mathbf{L}_0 = \log(F_0(\mathbf{x}))$ , as in (5.6), except with  $\pi_0$  known. The  $K \times K$  derivative matrix  $d\hat{\mathbf{L}}/d\hat{\mathbf{l}}$  has elements

$$\frac{\partial \hat{L}_j}{\partial \hat{l}_k} = \begin{cases} \hat{f}_k/\hat{F}_j & \text{if } k \geq j \\ 0 & \text{if } k < j. \end{cases} \quad (7.62)$$

Calling this matrix  $\hat{B}$ , the chain rule of differentiation gives

$$d\hat{\mathbf{L}}/d\mathbf{y} = \hat{B}M\hat{G}M' \quad (7.63)$$

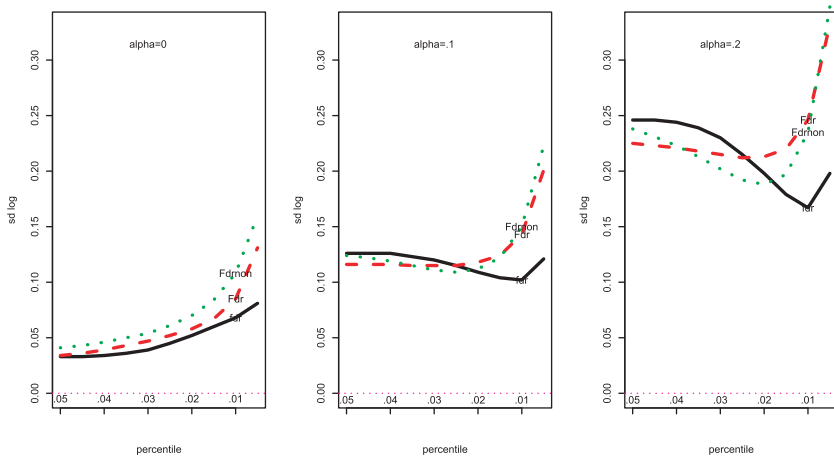
and

$$\widehat{\mathbf{Cov}}(\widehat{\mathbf{lfd}}r) = (\hat{B}M\hat{G}M') \widehat{\mathbf{Cov}}(\mathbf{y}) (M\hat{G}M' \hat{B}'). \quad (7.64)$$

**Exercise 7.7** Verify (7.62) and (7.64).

Figure 7.5 compares the accuracy of  $\widehat{\mathbf{fdr}}(z)$  with  $\widehat{\mathbf{Fdr}}(z)$  for the two-class model (5.18) with  $N = 6000$  and  $\alpha = 0, 0.1$ , or  $0.2$ . The solid curves graph the standard deviation of  $\log \widehat{\mathbf{fdr}}(z)$  obtained from (7.58), (7.49), plotted versus the right percentiles of the mixture distribution  $f(z)$ . The corresponding standard deviations for  $\log \widehat{\mathbf{Fdr}}(z)$ , obtained from (7.64), (7.50), appear as the dashed curves. Remembering that standard deviation on the log scale nearly equals the coefficient of variation, Exercise 5.3, we can see the CVs increasing from around 5% to over 20% as the correlation parameter  $\alpha$  goes from 0 to 0.2.

The dotted curves in Figure 7.2 depict the accuracy of the non-parametric



**Figure 7.5** Solid curves show standard deviation of  $\log \widehat{\text{fdr}}(z)$  as a function of  $z$  at the upper percentiles of  $f(z)$  for two-class model (5.18),  $N = 6000$ ,  $\alpha = 0, 0.1$ , and  $0.2$ ; dashed curves show same for  $\log \widehat{\text{Fdr}}(z)$ ; dotted curves denote non-parametric version  $\log \widehat{\text{Fdr}}(z)$ . Accuracy decreases sharply with increasing correlation  $\alpha$ . The local fdr estimate is more accurate at the extreme percentiles.

tail-area Fdr estimate  $\overline{\text{Fdr}}(z) = \pi_0 F_0(z) / \bar{F}(z)$  (7.1), obtained using (7.50) and

$$\frac{d \log \bar{F}_j}{dy_k} = \begin{cases} 1/(N\bar{F}_j) & \text{if } k \geq j \\ 0 & \text{if } k < j. \end{cases} \quad (7.65)$$

All three methods perform about equally well except at the extreme right, where  $\widehat{\text{fdr}}(z)$  is noticeably more accurate.

**Exercise 7.8** Verify (7.65).

All of our calculations have so far assumed that  $\pi_0$  and  $f_0(z)$  are known. Having  $\pi_0$  unknown barely degrades accuracy, but  $f_0$  is a different matter. Next we will calculate the influence function of  $\widehat{\text{fdr}}(z) = \hat{\pi}_0 \hat{f}_0(z) / \hat{f}(z)$  when  $\hat{\pi}_0$  and  $\hat{f}_0(z)$  are estimated by central matching, as in Figure 6.2.

Suppose that central matching is accomplished by a least squares quadratic fit of the counts  $y_k$  to the bin centers  $x_k$ , over a central subset of  $K_0$  bins having index set  $\mathbf{k}_0$ . Define  $M_0$  as the  $K \times 3$  matrix with  $k$ th row  $(1, x_k, x_k^2)$

and let  $\tilde{M}$  and  $\tilde{M}_0$  be the submatrices whose columns are defined by  $\mathbf{k}_0$ ,

$$\tilde{M} = M[\mathbf{k}_0] \quad \text{and} \quad \tilde{M}_0 = M_0[\mathbf{k}_0] \quad (7.66)$$

of dimensions  $K_0 \times m$  and  $K_0 \times 3$ .

**Theorem 7.4** *The influence function of  $\log \widehat{\text{fdr}}$  with respect to  $\mathbf{y}$  when using central matching is*

$$\frac{d\widehat{\text{lfdr}}}{d\mathbf{y}} = A\hat{G}M' \quad (7.67)$$

where

$$A = M_0\tilde{G}_0^{-1}\tilde{M}_0'\tilde{M} - M \quad \left[ \tilde{G}_0 = \tilde{M}_0'\tilde{M}_0 \right]. \quad (7.68)$$

Proof of Theorem 7.4 appears at the end of this section.

*Note* The theorem applies as stated to other models for the null: if  $f_0(z)$  is assumed known, with only  $\pi_0$  unknown, then  $M_0$  becomes the  $K \times 1$  matrix of 1's; for a cubic null, the  $k$ th row of  $M_0$  is  $(1, x_k, x_k^2, x_k^3)$ .

Theorem 7.4 yields the delta-method estimate

$$\widehat{\text{cov}}(\widehat{\text{lfdr}}) = (A\hat{G}M)\widehat{\text{cov}}(\mathbf{y})(M'\hat{G}A'). \quad (7.69)$$

As in (7.60), the independence estimate  $\widehat{\text{cov}}(\mathbf{y}) \doteq \text{diag}(\hat{\nu})$  gives the compact expression

$$\widehat{\text{cov}}(\widehat{\text{lfdr}}) \doteq A\hat{G}A'. \quad (7.70)$$

Comparing (7.67) with (7.57), the latter amounts to taking the first term in  $A$  (7.68) to be zero, which would be appropriate if we had an infinite amount of prior information on  $\pi_0$  and  $f_0(z)$ .

Table 7.3 compares  $\text{sd}\{\log \widehat{\text{fdr}}(z)\}$  using the theoretical null (7.58) versus the empirical null (7.70). The standard errors are *much* larger for the empirical null. Partly, however, that reflects a limitation of our analysis, which focuses on unconditional standard deviations. Some of the variability in the empirical null case is “good variability,” in that  $\widehat{\text{fdr}}(z)$  is correctly adjusting for the true false discovery proportion, as in Figure 6.4. The central matching estimates  $(\hat{\delta}_0, \hat{\sigma}_0)$  are acting as approximate ancillary statistics, to use classical terminology, which adapt the estimate  $\widehat{\text{fdr}}(z)$  to differing situations. At the present level of development, it is difficult to separate out the good and bad variability components of the empirical null estimate.

Table 7.3 Comparison of  $\text{sd}\{\log \widehat{\text{fdr}}(z)\}$  for  $N = 6000$ ,  $\alpha = 0.1$  (5.18), as in middle panel of Figure 7.5. Top row uses central matching (7.70); bottom row assumes  $\pi_0$ ,  $f_0(z)$  known (7.58). Some of the extra variability in the empirical case comes from correctly adjusting  $\widehat{\text{fdr}}(z)$  to differing situations, as in Figure 6.4.

| Percentile               | .05        | .04        | .03        | .02        | .01        |
|--------------------------|------------|------------|------------|------------|------------|
| <b>sd empirical null</b> | <b>.18</b> | <b>.26</b> | <b>.36</b> | <b>.54</b> | <b>.83</b> |
| sd theoretical null      | .13        | .13        | .12        | .11        | .10        |

*Proof of Theorem 7.4* In a Poisson GLM, the MLE of the expectation vector  $\boldsymbol{\nu} = \exp(M\boldsymbol{\beta})$  satisfies

$$M' [\mathbf{y} - e^{M\boldsymbol{\beta}}] = 0. \quad (7.71)$$

A small change  $d\mathbf{y}$  in  $\mathbf{y}$  therefore produces change  $d\hat{\boldsymbol{\beta}}$  satisfying

$$M' d\mathbf{y} = M' \text{diag}(\hat{\boldsymbol{\nu}}) M d\hat{\boldsymbol{\beta}}. \quad (7.72)$$

**Exercise 7.9** Verify (7.72) and use it to prove (7.57).

Let  $\hat{\mathbf{l}}_{00} = \log(\hat{\pi}_0 \hat{\mathbf{f}}_0)$ , the log numerator of  $\widehat{\text{fdr}}$ , which we are modeling in central matching by

$$\hat{\mathbf{l}}_{00} = M_0 \hat{\boldsymbol{\gamma}} \quad (7.73)$$

where  $\hat{\boldsymbol{\gamma}}$  is an  $m_0$ -dimensional vector and  $m_0 = 3$  as before (7.66). We are fitting  $\hat{\mathbf{l}}_{00}$  by least squares to the central portion  $\tilde{\mathbf{l}} = \hat{\mathbf{l}}[k_0]$  of  $\hat{\mathbf{l}} = \log \hat{\mathbf{f}}$ , so ordinary least squares theory gives

$$d\hat{\boldsymbol{\gamma}} = \tilde{G}_0^{-1} M'_0 d\tilde{\mathbf{l}} \quad \text{and} \quad d\hat{\mathbf{l}}_{00} = M_0 \tilde{G}_0^{-1} M'_0 d\tilde{\mathbf{l}}. \quad (7.74)$$

Then (7.57) yields

$$d\tilde{\mathbf{l}} = d\hat{\mathbf{l}}[k_0] = \tilde{M} \hat{G} M' d\mathbf{y} \quad (7.75)$$

and

$$d\hat{\mathbf{l}}_{00} = M_0 \tilde{G}_0^{-1} \tilde{M}' \tilde{M} \hat{G} M' d\mathbf{y} \quad (7.76)$$

from (7.74). Finally,

$$d\widehat{\text{fdr}} = d(\hat{\mathbf{l}}_{00} - \tilde{\mathbf{l}}) = (M_0 \hat{G}_0 \tilde{M}' \tilde{M} - M) \hat{G} M' d\mathbf{y}, \quad (7.77)$$

verifying (7.67).  $\square$

### 7.4 The Non-Null Distribution of $z$ -Values

The main results of this chapter, summarized in (7.49)–(7.50), depend on normality,  $z_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$  (7.3). In all of our examples, the  $z_i$  were actually  $z$ -values, i.e., standardized test statistics, obtained for instance as in (2.2)–(2.6). By definition, a  $z$ -value is a statistic having a  $\mathcal{N}(0, 1)$  distribution under a null hypothesis  $H_0$  of interest. But will it still be normal under non-null conditions? In other words, is it legitimate to apply results like (7.2) to the leukemia data, as we did in Figure 7.1?

The results of this section are reassuring. They show that, under repeated sampling, the non-null distribution of a typical  $z$ -value will have standard deviation of order  $O(n^{-1/2})$  and non-normality  $O_p(n^{-1})$  as we move away from the null hypothesis: so  $z_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$  holds to a good approximation, though we can expect  $\sigma_i^2$  to differ noticeably from 1 when  $\mu_i$  is far from 0.

Figure 7.6 illustrates the phenomenon for the Student- $t$  situation (2.2)–(2.6). Here we have

$$z = \Phi^{-1}(F_\nu(t)) \quad \text{with } t \sim t_\nu(\delta), \quad (7.78)$$

the notation  $t_\nu(\delta)$  indicating a non-central  $t$  variable with  $\nu$  degrees of freedom and non-centrality parameter  $\delta$  (not  $\delta^2$ ),

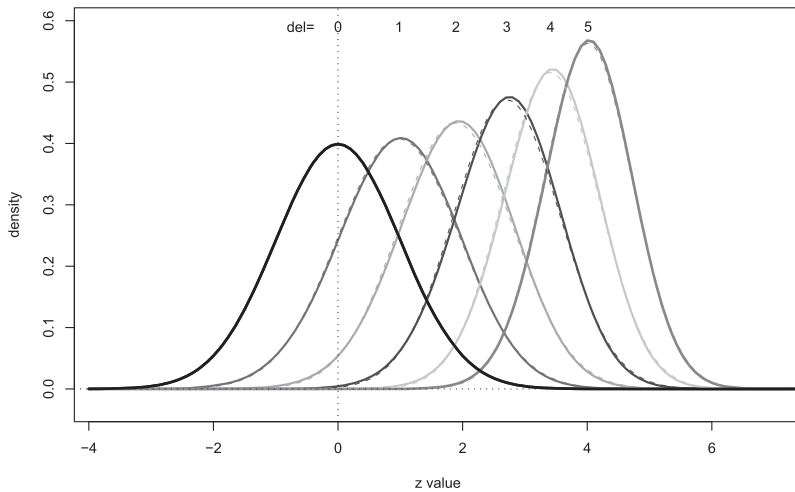
$$t = \frac{Z + \delta}{S} \quad \left[ Z \sim \mathcal{N}(0, 1) \text{ independent of } S^2 \sim \chi_\nu^2/\nu \right]. \quad (7.79)$$

$F_\nu$  is the cdf of a *central*  $t_\nu$  distribution as in (2.5), where  $\nu = 100$ .

Table 7.4 *Non-central  $t$  example  $t \sim t_\nu(\delta)$  for  $\nu = 20$ ,  $\delta = 0, 1, 2, 3, 4, 5$ ; moment parameters of  $z = \Phi^{-1}(F_\nu(t))$  (7.78) indicate near-normality even for  $\delta$  far from 0.*

|      | $\delta$ | 0    | 1    | 2    | 3    | 4    | 5 |
|------|----------|------|------|------|------|------|---|
| mean | 0        | .98  | 1.89 | 2.71 | 3.41 | 4.01 |   |
| sd   | 1        | .98  | .92  | .85  | .77  | .71  |   |
| skew | 0        | -.07 | -.11 | -.11 | -.10 | -.07 |   |
| kurt | 0        | .02  | .06  | .08  | .09  | .07  |   |

Figure 7.6 shows the density of  $z$  from (7.78) for  $\nu = 20$  and non-centrality  $\delta = 0, 1, 2, 3, 4, 5$ . By definition, the density is  $\mathcal{N}(0, 1)$  for  $\delta = 0$ . At  $\delta = 5$ , the density almost exactly matches that of a  $\mathcal{N}(4.01, 0.71^2)$  distribution. Mean, standard deviation, skewness, and kurtosis for each choice of  $\delta$  are given in Table 7.4, bolstering faith in the approximation  $z \sim \mathcal{N}(\mu, \sigma^2)$ .



**Figure 7.6**  $z$ -value density for non-central  $t$  distribution (7.78)–(7.79), for degrees of freedom  $\nu = 20$  and non-centrality  $\delta = 0, 1, 2, 3, 4, 5$ . Dashed curves are normal densities matched in mean and standard deviation; see Table 7.4. Negative  $\delta$  gives mirror image results.

**Exercise 7.10** A test statistic  $t$  has possible densities  $\{f_\theta(t), \theta \in \Theta\}$  with corresponding cdfs  $F_\theta(t)$  and we wish to test  $H_0 : \theta = \theta_0$ . Show that the  $z$ -value statistic  $z = \Phi^{-1}\{F_{\theta_0}(t)\}$  has densities

$$g_\theta(z) = \varphi(z)f_\theta(t)/f_{\theta_0}(t) \quad (7.80)$$

with  $\varphi(z)$  the standard normal density (2.9).

$T$ -statistics are ubiquitous in statistical applications. Suppose that in addition to the  $N \times n$  expression matrix  $\mathbf{X}$  in a microarray experiment, we observe a primary response variable  $u_j$  ( $u_j$  the treatment/control indicator in a two-sample study) and covariates  $w_{j1}, w_{j2}, \dots, w_{jp}$  for the  $j$ th of  $n$  subjects. Given the observed expression levels  $x_{i1}, x_{i2}, \dots, x_{in}$  for gene  $i$ , we could calculate  $t_i$ , the usual  $t$ -value for  $u_j$  as a function of  $x_{ij}$  after adjusting for the covariates in a linear model. Then, as in (7.78),

$$z_i = \Phi^{-1}\left(F_{n-p-1}(t_i)\right) \quad (7.81)$$

would be a  $z$ -value under the usual Gaussian assumption, with behavior like that in Table 7.4.

As a second example where exact calculations are possible, suppose we

observe an i.i.d. sample  $x_1, x_2, \dots, x_n$  from an exponential distribution of unknown scale  $\theta$ ,

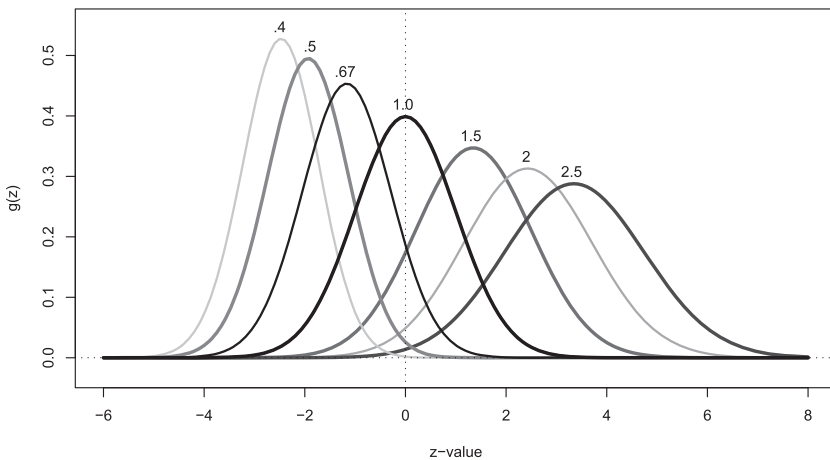
$$f_{\theta}(x) = \frac{1}{\theta} e^{-x/\theta} \quad [x \geq 0]. \quad (7.82)$$

An exact  $z$ -value  $Z_0$  for testing  $H_0 : \theta = \theta_0$  is based on the mean  $\bar{x}$ ,

$$Z_0 = \Phi^{-1}(G_n(n\bar{x}/\theta_0)) \quad (7.83)$$

where  $G_n$  is the cdf of a gamma distribution with  $n$  degrees of freedom.

**Exercise 7.11** Using the fact that  $\bar{x}$  is distributed as  $\theta/n$  times a gamma variable with  $n$  degrees of freedom, show that  $Z_0 \sim \mathcal{N}(0, 1)$  if  $\theta = \theta_0$ .



**Figure 7.7**  $z$ -value density curves for gamma example,  $Z_0$  as in (7.83) with  $n = 10$ ,  $\theta_0 = 1$ . True value  $\theta$  indicated above curves. Normality for non-null cases is nearly perfect here; see Table 7.5.

Figure 7.7 displays the density of  $Z_0$  for different choices of  $\theta$  when  $n = 10$  and  $\theta_0 = 1$ . Normality is excellent: there is no visible discrepancy between the density curves and their matching normal equivalents (having means and standard deviations as in Table 7.5).

What follows is a heuristic argument showing why we can expect non-null normality for  $z$ -values obtained from an independent and identically distributed (i.i.d.) sample.<sup>7</sup> Suppose that  $x_1, x_2, \dots, x_n$  is such a sample,

<sup>7</sup> The argument is more technical than the previous material, but yields a simple and useful result at (7.102).

Table 7.5 Mean, standard deviation, skewness, and kurtosis for  $Z_0$  (7.83); the gamma example with  $n = 10$  and  $\theta_0 = 1$ . For true  $\theta$  in (7.82) as indicated.

| $\theta$ | .4    | .5    | .67   | 1 | 1.5  | 2.0  | 2.5  |
|----------|-------|-------|-------|---|------|------|------|
| mean     | -2.49 | -1.94 | -1.19 | 0 | 1.36 | 2.45 | 3.38 |
| stdev    | .76   | .81   | .88   | 1 | 1.15 | 1.27 | 1.38 |
| skew     | -.05  | -.04  | -.02  | 0 | .02  | .04  | .04  |
| kurt     | .01   | .01   | .00   | 0 | .00  | -.01 | -.04 |

obtained from cdf  $F_\theta$ , a member of a one-parameter family

$$\mathcal{F} = \{F_\theta, \theta \in \Theta\} \quad (7.84)$$

having its moment parameters {mean, standard deviation, skewness, kurtosis}, denoted by

$$\{\mu_\theta, \sigma_\theta, \gamma_\theta, \delta_\theta\}, \quad (7.85)$$

defined differentially in  $\theta$ .

Under the null hypothesis  $H_0 : \theta = 0$ , which we can write as

$$H_0 : x \sim \{\mu_0, \sigma_0, \gamma_0, \delta_0\} \quad (7.86)$$

the standardized variate

$$Y_0 = \sqrt{n} \left( \frac{\bar{x} - \mu_0}{\sigma_0} \right) \quad \left[ \bar{x} = \sum_{i=1}^n y_i/n \right] \quad (7.87)$$

satisfies

$$H_0 : Y_0 \sim \left\{ 0, 1, \frac{\gamma_0}{\sqrt{n}}, \frac{\delta_0}{n} \right\}. \quad (7.88)$$

Normality can be improved to second order by means of a Cornish–Fisher transformation,

$$Z_0 = Y_0 - \frac{\gamma_0}{6\sqrt{n}} (Y_0^2 - 1) \quad (7.89)$$

which reduces the skewness in (7.88) from  $O(n^{-1/2})$  to  $O(n^{-1})$ ,

$$H_0 : Z_0 \sim \{0, 1, 0, 0\} + O(n^{-1}). \quad (7.90)$$

We can interpret (7.90) as saying that  $Z_0$  is a *second-order  $z$ -value*,

$$H_0 : Z_0 \sim \mathcal{N}(0, 1) + O_p(n^{-1}) \quad (7.91)$$

e.g., a test statistic giving standard normal  $p$ -values accurate to  $O(n^{-1})$ .

Suppose now that  $H_0$  is false and instead  $H_1$  is true, with  $x_1, x_2, \dots, x_n$  i.i.d. according to

$$H_1 : x \sim \{\mu_1, \sigma_1, \gamma_1, \delta_1\} \quad (7.92)$$

rather than (7.86). Setting

$$Y_1 = \sqrt{n} \left( \frac{\bar{x} - \mu_1}{\sigma_1} \right) \quad \text{and} \quad Z_1 = Y_1 - \frac{\gamma_1}{6\sqrt{n}} (Y_1^2 - 1) \quad (7.93)$$

makes  $Z_1$  second-order normal under  $H_1$ ,

$$H_1 : Z_1 \sim \mathcal{N}(0, 1) + O_p(n^{-1}). \quad (7.94)$$

We wish to calculate the distribution of  $Z_0$  (7.89) under  $H_1$ . Define

$$c = \sigma_1/\sigma_0, \quad d = \sqrt{n}(\mu_1 - \mu_0)/\sigma_0, \quad \text{and} \quad g_0 = \gamma_0 / (6\sqrt{n}). \quad (7.95)$$

Some simple algebra yields the following relationship between  $Z_0$  and  $Z_1$ .

**Lemma 7.5** Under definitions (7.89), (7.93), and (7.95),

$$Z_0 = M + SZ_1 + g_0 \left\{ \left( \frac{\gamma_1}{\gamma_0} S - c^2 \right) (Y_1^2 - 1) + (1 - c^2) \right\} \quad (7.96)$$

where

$$M = d \cdot (1 - dg_0) \quad \text{and} \quad S = c \cdot (1 - 2dg_0). \quad (7.97)$$

The asymptotic relationships claimed at the start of this section are easily derived from Lemma 7.5. We consider a sequence of alternatives  $\theta_n$  approaching the null hypothesis value  $\theta_0$  at rate  $n^{-1/2}$ ,

$$\theta_n - \theta_0 = O(n^{-1/2}). \quad (7.98)$$

The parameter  $d = \sqrt{n}(\mu_{\theta_n} - \mu_0)/\sigma_0$  defined in (7.95) is then of order  $O(1)$ , as is

$$M = d(1 - dg_0) = d \left( 1 - d\gamma_0 / (6\sqrt{n}) \right) \quad (7.99)$$

while standard Taylor series calculations give

$$c = 1 + \frac{\dot{\sigma}_0}{\dot{\mu}_0} \frac{d}{\sqrt{n}} + O(n^{-1}) \quad \text{and} \quad S = 1 + \left( \frac{\dot{\sigma}_0}{\dot{\mu}_0} - \frac{\gamma_0}{3} \right) \frac{d}{\sqrt{n}} + O(n^{-1}), \quad (7.100)$$

the dot indicating differentiation with respect to  $\theta$ .

**Theorem 7.6** Under model (7.84), (7.98), and the assumptions of Lemma 7.5,

$$Z_0 \sim \mathcal{N}(M, S^2) + O_p(n^{-1}) \quad (7.101)$$

with  $M$  and  $S$  as given in (7.99)–(7.100). Moreover,

$$\left. \frac{dS}{dM} \right|_{\theta_0} = \frac{1}{\sqrt{n}} \left( \left. \frac{d\sigma}{d\mu} \right|_{\theta_0} - \frac{\gamma_0}{3} \right) + O(n^{-1}). \quad (7.102)$$

*Proof* The proof of Theorem 7.6 uses Lemma 7.5, with  $\theta_n$  determining  $H_1$  in (7.96). Both  $1 - c^2$  and  $(\gamma_1/\gamma_0)S - c^2$  are of order  $O(n^{-1/2})$ ; the former from (7.100) and the latter using  $\gamma_1/\gamma_0 = 1 + (\dot{\gamma}_0/\gamma_0)(\theta_n - \theta_0) + O(n^{-1})$ . Since  $Y_1^2 - 1$  is  $O_p(1)$ , this makes the bracketed term in (7.96)  $O_p(n^{-1/2})$ ; multiplying by  $g_0 = \gamma_0/(6\sqrt{n})$  reduces it to  $O_p(n^{-1})$ , and (7.101) follows from (7.94). Differentiating  $M$  and  $S$  in (7.99)–(7.100) with respect to  $d$  verifies (7.102).  $\square$

Theorem 7.6 supports our claim that, under non-null alternatives, the null hypothesis normality of  $Z_0$  degrades more slowly than its unit standard deviation, the comparison being  $O_p(n^{-1})$  versus  $O(n^{-1/2})$ .

Result (7.102) simplifies further if  $\mathcal{F}$  is a one-parameter exponential family, i.e., a family with densities proportional to  $\exp\{\theta x\} \cdot f_0(x)$ . Then  $d\sigma/d\mu = \gamma/2$ , so that  $(M, S)$  in (7.101) satisfy

$$\left. \frac{dS}{dM} \right|_{\theta_0} = \frac{\gamma_0}{6\sqrt{n}} + O(n^{-1}). \quad (7.103)$$

For the gamma family,  $\gamma_0 = 2$ , giving  $\gamma_0/(6\sqrt{n}) = 0.1054$  for  $n = 10$ . This matches to better than three decimal places the observed change in standard deviation with respect to mean, as illustrated in Figure 7.7.

Moving beyond one-parameter families, suppose  $\mathcal{F}$  is a  $p$ -parameter exponential family, having densities proportional to  $\exp\{\eta_1 x_1 + \eta_2' x_2\} f_0(x_1, x_2)$ , where  $\eta_1$  and  $x_1$  are real-valued while  $\eta_2$  and  $x_2$  are  $(p-1)$ -dimensional vectors, but where we are only interested in  $\eta_1$  and not the nuisance vector  $\eta_2$ . The conditional distribution of  $x_1$  given  $x_2$  is then a one-parameter exponential family with natural parameter  $\eta_1$ , which puts us back in the context of Theorem 7.6.

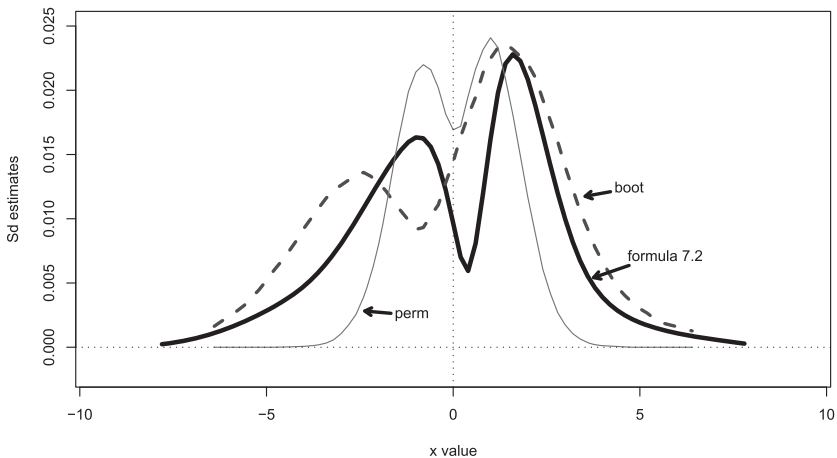
The non-central  $t$  family does not meet the conditions of Lemma 7.5 or Theorem 7.6: (7.78) is symmetric in  $\delta$  around zero, causing  $\gamma_0$  in (7.98) to equal 0 and likewise the derivative in (7.102). Nevertheless, as Figure 7.5 shows, it does exhibit impressive non-null normality. In fact, my experience has been that  $z$ -values in general follow near-normal non-null distributions,

making the crucial assumption (7.3) a reasonable starting point for their analysis.

## 7.5 Bootstrap Methods

It would be very nice to avoid all the normal theory parametric intricacies of the past sections, and obtain our accuracy estimates directly from a non-parametric bootstrap. Figure 7.8 tries this for the situation illustrated in Figure 7.1. Here the data matrix  $X$  has  $N = 7128$  rows (genes) and  $n = 72$  columns (patients), with  $n_1 = 45$  ALL and  $n_2 = 25$  AML patients as in Figure 6.1a.

Bootstrap data sets  $X^*$  were obtained by randomly selecting 45 of the first 45  $X$  columns *with replacement*, and likewise 27 of the last 27 columns (under the assumption that the patients, though not the genes, were sampled independently).  $X^*$  gave  $N$ -vector  $z^*$  as for the actual data, and then  $\bar{F}(x)^*$  (7.1).  $B = 50$  replications of this whole process yielded values  $\bar{F}(x)^{*1}, \bar{F}(x)^{*2}, \dots, \bar{F}(x)^{*50}$ , whose empirical standard deviation  $\widehat{\text{sd}}(x)$  was the non-parametric bootstrap standard error estimate for  $\bar{F}(x)$ .



**Figure 7.8** Standard deviation estimates for leukemia data cdf as in Figure 7.1. Non-parametric bootstrap (dashed line) gives a dilated version of  $\text{sd}(x)$  from parametric formula (7.2).

Permutation  $\widehat{\text{sd}}(x)$  curve (thin line) is too narrow.

The dashed curve in Figure 7.8 traces  $\widehat{\text{sd}}(x)$  for  $-8 < x < 8$ . It looks like a dilated version of the curve from (7.2), with both peaks noticeably

widened. This turns out to be an unfortunately dependable feature of non-parametric bootstrapping in an  $N \gg n$  situation: for each  $i$ , the bootstrap replications  $z_i^{*1}, z_i^{*2}, \dots, z_i^{*50}$  roughly follow a normal distribution centered at  $z_i$ , say

$$z_i^* \sim \mathcal{N}(z_i, \hat{v}_i) \quad i = 1, 2, \dots, N. \quad (7.104)$$

This widens the original  $z$ -value histogram of Figure 6.1a by adding the  $\hat{v}_i$  component of variance, and subsequently widening the  $\widehat{\text{sd}}(x)$  curve.

Figure 7.8 also shows a permutation  $\widehat{\text{sd}}(x)$  curve, where now the  $z^*$  vectors are obtained by permuting all 72 columns of  $X$ . There are two reasons that this curve is much narrower than that from formula (7.2). The less important one here is that permutations nullify any true differences between ALL and AML patients (in contrast to the two-sample bootstrapping above). More crucially, permutations tend to enforce a  $\mathcal{N}(0, 1)$  null distribution, Figure 6.5, much narrower than the  $\mathcal{N}(0.09, 1.68^2)$  null seen in Figure 6.1a.

Column-wise bootstrapping and permuting both preserve within-column correlations. (Their correctly bimodal  $\widehat{\text{sd}}(x)$  curves in Figure 7.8 reflect this.) Trouble for the non-parametric bootstrap stems from a simpler cause: the empirical distribution of  $n$  points in  $N$  dimensions is a poor estimator of the true distribution when  $N$  is much larger than  $n$ . Parametric bootstrapping is less prone to failure, but of course that assumes knowledge of a reasonably accurate parametric model for  $X$ .

## Notes

Most of this chapter's material is taken from Efron (2007a) and Efron (2010). Other good references for large-scale correlation effects include Dudoit et al. (2004); Owen (2005); Qiu et al. (2005a,b); Clarke and Hall (2009); and Desai et al. (2010). Schwartzman and Lin (2009) use Mehler's expansion to pursue higher-order accuracy formulas. The influence function calculations of Section 7.3 appear in Efron (2007b, Sect. 5) as part of a more extensive array of accuracy formulas.

Lancaster (1958) provides an insightful discussion of Mehler's formula, published in 1866! There are a variety of sources for Cornish–Fisher expansions, including Chapter 2 of Johnson and Kotz (1970) and, for much greater detail, Section 2.2 of Hall (1992).

The quantity  $\gamma_0/6\sqrt{n}$  (7.103) is called the *acceleration* in Efron (1987), interpreted as “the rate of change of standard deviation with respect to

expectation on a normalized scale,” which agrees nicely with its use in (7.103).

Efron and Tibshirani (1993) provide a full treatment of the bootstrap, with emphasis on the importance of estimating the underlying probability distribution of the sampling units, as in their Figure 8.3. Effective bootstrapping usually requires  $n \gg N$ , just what we don’t have here.